

DUKOVANY, Czech Republic

WMO: 116930

Lat: 49.0956N

Lon: 16.1344E

Elev: 1319

StdP: 14.01

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.4	14.9	3.9	7.0	12.8	9.0	9.1	17.2	32.4	39.7	29.5	37.3	7.9	350	0.607

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	86.9	67.0	83.5	65.8	80.3	64.5	69.0	81.9	67.4	79.7	66.0	77.2	9.8	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.8	96.5	73.3	63.3	91.4	71.6	61.9	86.8	70.4	33.9	82.0	32.6	79.9	31.4	77.4	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.5	23.0	20.2	6.1	91.7	5.4	3.3	2.3	94.1	-0.9	96.0	-3.9	97.9	-7.8	100.3
			5.4	72.1	5.2	1.7	1.6	73.3	-1.5	74.3	-4.4	75.3	-8.2	76.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.3	29.2	32.3	39.2	48.9	56.6	62.9	66.9	66.5	57.8	48.4	39.4	30.9	(6)
	DBStd	15.09	7.79	8.28	7.51	7.69	6.57	6.60	6.10	6.10	6.28	6.93	7.21	7.42	(7)
	HDD50	2650	646	495	339	111	17	1	0	0	9	117	324	591	(8)
	HDD65	6371	1111	915	798	483	270	118	53	55	228	516	768	1056	(9)
	CDD50	2040	0	0	6	78	221	387	523	511	241	67	6	0	(10)
	CDD65	285	0	0	0	0	9	54	110	101	11	0	0	0	(11)
	CDH74	2678	0	0	0	8	92	517	1064	907	89	1	0	0	(12)
	CDH80	751	0	0	0	0	11	121	341	269	9	0	0	0	(13)
(14) Wind	WSAvg	9.4	9.7	10.0	10.4	10.1	9.6	8.8	9.1	8.4	9.0	8.8	9.2	9.2	(14)
(15) Precipitation	PrecAvg	20.3	0.9	0.8	1.3	1.3	2.3	2.6	2.9	2.4	1.9	1.3	1.3	1.1	(15)
	PrecMax	30.4	1.9	2.3	3.5	2.8	4.9	5.7	7.8	5.4	4.6	2.7	2.5	1.7	(16)
	PrecMin	15.4	0.1	0.1	0.3	0.1	0.6	0.6	0.5	0.9	0.3	0.2	0.1	0.3	(17)
	PrecStd	3.4	0.5	0.6	0.8	0.7	0.9	1.2	1.5	1.2	1.2	0.7	0.6	0.4	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.2	54.0	63.9	74.7	81.0	88.2	91.6	91.3	81.0	70.2	59.7	49.5	(19)
		MCWB	46.1	46.6	50.6	58.0	63.5	67.9	67.8	67.0	63.7	58.7	53.8	46.8	(20)
	2%	DB	46.0	49.8	59.3	70.1	76.4	83.6	87.3	86.5	76.4	65.3	55.4	46.1	(21)
		MCWB	42.5	44.3	48.5	55.8	62.0	67.3	67.0	66.5	62.0	57.1	51.2	43.4	(22)
	5%	DB	42.3	46.6	54.7	66.2	72.9	80.0	83.9	82.6	72.6	61.9	51.9	43.0	(23)
		MCWB	39.6	42.2	46.1	53.1	60.1	64.9	65.7	65.2	60.5	55.5	48.9	40.8	(24)
	10%	DB	39.1	43.1	50.7	62.4	69.4	76.4	80.2	79.1	69.0	58.6	49.0	40.2	(25)
		MCWB	37.1	39.7	44.4	51.3	58.1	63.4	64.5	63.9	59.3	54.2	46.6	38.6	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.1	47.7	52.8	59.8	66.2	71.4	71.4	70.1	66.7	60.8	54.6	47.1	(27)
		MCDB	51.1	53.0	60.9	72.0	76.7	83.1	83.6	83.8	75.6	66.4	58.6	49.3	(28)
	2%	WB	43.1	45.2	49.5	56.8	63.5	68.8	68.9	68.3	64.1	58.3	51.9	43.8	(29)
		MCDB	45.8	49.0	56.7	68.2	73.9	80.0	82.2	82.0	72.8	62.9	55.0	45.8	(30)
	5%	WB	39.9	42.5	47.3	54.2	61.4	66.5	67.3	66.6	62.0	56.4	49.2	41.1	(31)
		MCDB	41.9	46.0	53.5	63.8	70.5	76.7	80.0	79.2	69.9	60.6	51.6	42.7	(32)
	10%	WB	37.2	39.9	45.0	52.2	59.2	64.6	65.7	65.2	60.2	54.5	46.9	38.9	(33)
		MCDB	38.9	42.8	49.9	60.6	67.5	73.9	77.2	76.3	67.6	58.3	48.8	40.2	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	8.2	11.0	14.0	17.8	18.0	19.3	20.0	19.2	15.9	12.5	8.6	7.6	(35)
		MCDBR	11.9	15.4	21.4	23.6	23.2	24.5	25.9	24.4	21.7	17.5	12.5	10.9	(36)
	5% WB	MCWBR	9.6	10.7	12.5	11.7	11.5	10.5	9.2	9.0	9.9	10.1	8.7	8.9	(37)
		MCWBR	11.5	14.4	19.2	21.5	21.4	22.3	23.1	21.9	18.9	15.3	11.7	10.2	(38)
(40) Clear-Sky Solar Irradiance	taub		0.303	0.326	0.365	0.409	0.410	0.411	0.421	0.418	0.389	0.361	0.337	0.302	(40)
		taud	2.361	2.341	2.284	2.200	2.238	2.266	2.241	2.266	2.318	2.367	2.385	2.401	(41)
	Ebn at Noon		238	259	266	265	270	270	265	260	255	242	222	222	(42)
		Edh at Noon	22	28	35	42	42	41	42	39	33	27	21	18	(43)
(44) All-Sky Solar Radiation	RadAvg		288	552	900	1366	1613	1751	1675	1497	1063	627	311	223	(44)
	RadStd		46	73	137	159	187	146	171	143	142	85	34	24	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (75 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		+0.95	N/A	N/A	N/A	+0.70	+0.59	-221	-288	+126	+50

Nomenclature: See separate page